

Faculty Seminar on the Foundations of Data Science

Term: Fall 2026

Schedule: Bi-weekly, times TBD

Location: In-person, location TBD

An informal seminar among faculty in the School of Data Science on the foundations of data science. The objective of the seminar is to develop and codify a shared understanding of data science imagined as a unified discipline. The method for reaching this objective is a series of conversations focused on key texts in the theory and history of data science.

Schedule

The seminar will take place by

Theoretical Takes

Week 1. Data Science and Closure

Bieling, et al., 2026, “Closure and the Foundations of Data Science.”

Week 2: The 4+1 Model Revisited

Alvarado, 2023, “The 4+1 Model of Data Science.”

<https://arxiv.org/abs/2311.07631>

Week 3: Data Science and the Scientific Method

Wright, Alonzi, et al., 2026, “The Importance of the Scientific Method in Machine Learning Education.”

<https://hdr.mitpress.mit.edu/pub/xxb041qb/release/1>

Historical Contexts

Week 4: The First Data Scientist

1. The First Data Scientists
 1. AFCRL Report
 2. Peter Naur’s Dataology
2. Scientists
 1. CODATA
 2. Data Science Journal
 3. NSF and JISC reports
3. Statisticians
 1. The Tokyo School
 2. Cleveland, et al.
 3. Breiman